

Expertise Coordination in Information Systems Development Projects: Willingness, Ability, and Behavior

Jack Shih-Chieh Hsu, *National Sun Yat-sen University, Taiwan, R.O.C.*

Yu Wen Hung, *National Sun Yat-sen University, Taiwan, R.O.C.*

Sheng-Pao Shih, *Tamkang University, Taiwan, R.O.C.*

Hui-Mei Hsu, *National Kaohsiung Normal University, Taiwan, R.O.C.*

ABSTRACT ■

Information systems development (ISD) projects are complex, requiring a variety of expertise. Coordinating such expertise helps manage complexity, increasing the likelihood of a project's success. Findings of past studies have been inconsistent regarding the benefits of expertise coordination—perhaps, in part, because three different forms of coordination have been used: willingness, ability, and behavior. We find that willingness and ability are antecedents of coordination behavior, and that coordination behavior fully mediates different forms of project success. Thus, successful expertise coordination requires team members who are both willing and able. The implications and limitations of this study are discussed.

KEYWORDS: coordination; information systems development (ISD) projects; project management; expertise; willingness; ability; organizational information processing theory

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INTRODUCTION ■

Information systems development (ISD) work is regarded as the crafting of an artifact to support organizational processes and provide information, communication, and processing support for various users across many disciplines. A variety of disciplines and interests is also represented in the project (Alexander & Robertson, 2004). Stakeholders usually include users, owners, and developers. Project teams are composed of individuals from technical, managerial, and operational backgrounds. Each stakeholder has a distinct set of prior expectations and criteria for the project, and each views success differently (Cao & Hoffman, 2011; Yu, Flett, & Bowers, 2005). In addition, because many information systems are large in scale, ISD projects bring about a high level of interdependence among individual workers (Kraut & Streeter, 1995). However, team members with different skills and expertise often have limited experience working together. In ISD projects, dependencies may exist among tasks, subtasks, and resources (Crowston & Kammerer, 1998). These confounding factors are believed to be major contributors to the high failure rate of ISD projects (Ibrahim, Ayazi, Nasrmaalek, & Nakhat, 2013; Lee, Park, & Lee, 2015; Narayanaswamy, Grover, & Henry, 2013; Wallace, Keil, & Rai, 2004a). ISD project teams, in particular, face the challenge of effectively coordinating the expertise of multiple stakeholders.

Experts agree that for an ISD project to be successful, all of the diverse talents, goals, and interests need to be effectively coordinated (Andres & Zmud, 2002; Banker, Bardhan, & Asdemir, 2006; Hoegl & Gemuenden, 2001; Hsu, Shih, Chiang, & Liu, 2012; Kraut & Streeter, 1995; Nidumolu, 1995; Reich, Gemino, & Sauer, 2008). One of the ISD team's more formative tasks is to coordinate the skills, talents, and knowledge required to enhance team performance or team effectiveness (Faraj & Sproull, 2000; Kanawattanachai & Yoo, 2007; Yang, Freeman, & Lynch, 2008). Given the value that effective coordination is expected to bring, prior studies have examined how different coordination mechanisms affect ISD projects (Andres & Zmud, 2002; Kirsch, 2000). Expertise coordination, in particular, is a critical factor in successful ISD projects (Faraj & Sproull, 2000; Okhuysen, 2001; Reich & Benbasat, 1996; Tiwana & McLean, 2005).

However, the findings of empirical studies in the IS literature have been inconsistent. For example, expertise coordination strongly influences project performance, team effectiveness, and team efficiency in software development projects (Faraj & Sproull, 2000; Jiang, Klein, & Chen, 2006). In contrast, the impacts of task-knowledge coordination on the performance of a virtual team only occurred at the end of 8-week project (Kanawattanachai & Yoo, 2007).

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The inconclusive results regarding the relationship between coordination and performance might be caused by the measures of coordination. Prior IS studies have tended to mix measures of coordination willingness, coordination ability, and coordination behavior (Collins & Smith, 2006). Coordination ability refers to the extent to which project team members are able to manage interdependencies, whereas coordination willingness refers to how much project team members are willing to exchange, combine, and integrate their unique expertise or knowledge to solve a problem. Factors that describe team members' cognitive, motivational, and affective states (e.g., willingness, cognitive ability) should be distinguished from team coordination behavior and activities (Marks, Mathieu, & Zaccaro, 2001). Constructs are contaminated when behavior and motivational or affective states are mixed together (Collins & Smith, 2006). Such ambiguous measurements tend to weaken the relationships between coordination and its antecedents and consequences (Gerwin, 2004). We propose three expertise coordination components: ability, willingness, and behavior. Thus, our first research question is: How are expertise coordination ability, expertise coordination willingness, and expertise coordination behavior interrelated?

More specifically, based upon the theory of planned behavior (Ajzen & Fishbein, 1980), we propose that expertise coordination behavior is affected by both the cognitive and affective factors of coordination, and that effective coordination behavior is an important antecedent of the success of an ISD project (Strode, Huff, Hope, & Link, 2012). For completeness, we model all three coordination components and consider success (i.e., project performance, system quality, and personal work satisfaction) from the viewpoints of various stakeholders. Thus, our second research question is: What impact does effective coordination behavior have on the success of an ISD project?

Overall, this study aims to investigate the impact of expertise coordination on ISD projects. Expertise coordination is broken down into three components: ability, willingness, and behavior. Furthermore, we examine the relationships among these three dimensions. We collected our data via a survey of 525 team members from 104 project teams. The results confirm the model and demonstrate how distinguishing coordination components is important for future research studying the effects of coordination behavior on multiple success indicators. This study provides insights for the impact of coordination on ISD projects and helps researchers understand how subdividing coordination has caused inconsistent results across past studies.

Background

Malone and Crowston (1990) defined coordination theory as a body of principles about how activities can be coordinated—that is, about how actors can work together harmoniously. Coordination among team members represents the act of working together in harmony as the team deals with a series of problems. This coordination is necessary for the team's daily activities, such as the subdivision of goals into actions, assigning actions to actors, resource allocation among different actors, and information sharing among actors to help achieve the overall goals. Organization theory describes coordination as the act of managing interdependencies between activities that are performed to achieve a common goal. These interdependencies include shared resources, prerequisites and usability constraints, simultaneous constraints, and task/subtask relationships (Malone & Crowston, 1994). Specifically, Van de Ven, Delbecq, and Koenig (1976) defined coordination as “*the integration of different parts of an organization to accomplish a collective set of tasks*” (p. 322) and argued that the impact of coordination on the project outcome depends on how effectively the team performs coordination behavior.

Many characteristics of ISD projects require team members to coordinate in order to solve problems and create the final product. First, many information systems are large in scale and cannot be implemented or even understood by a single small group (Hsu, Hung, Chen, & Huang, 2013). Second, both the rapidly changing business environment and the increasingly rapid rate of technological innovations create an inherent uncertainty to which the team must respond effectively (Shih, Shaw, Fu, & Cheng, 2013). Third, ISD projects usually involve originality and creativity, so the content of the project is most likely new to the team members (Kwak & Stoddard, 2004; Reich et al., 2008). Fourth, ISD projects still suffer from a high rate of failure (Ibrahim et al., 2013; Narayanaswamy et al., 2013). Research conducted by McKinsey and the University of Oxford reported that 45% of large IT projects run over budget, 7% run over time, and 56% deliver less value than expected (Bloch, Blumberg, & Laartz, 2012). Lastly, ISD projects lead to a high level of interdependence among individual workers (Kraut & Streeter, 1995), and team members with different levels of skill and expertise often have limited experience working together. In ISD projects, dependencies may exist between subtasks, between tasks and resources, and between resources (Crowston & Kammerer, 1998). These conditions make ISD projects a unique environment worthy of independent study, and also highlight the importance of coordination in such projects.

Coordination can be viewed as the management of interdependencies (Van de Ven et al., 1976). Effective coordination is particularly critical to the success of a software project (Strode et al., 2012). The degree of effective coordination in a project is determined by how well team members can manage the interdependencies (Faraj & Sproull, 2000). Effective coordination allows the work to be completed efficiently, without redundancy (Kraut & Streeter, 1995). It can also enhance the working climate

between users and IS developers (Li, Jiang, & Klein, 2003). Inter-function and inter-organization supply chain integration can also be enhanced by coordination efforts (Bharadwaj, Bharadwaj, & Bendoly, 2007). With sufficient information exchange, team members can cope with environmental uncertainty and variety (Nidumolu, 1995, 1996a). Expertise coordination is generally believed to serve as an important factor for creative and successful system development (Faraj & Sproull, 2000; Kanawattanachai & Yoo, 2007; Mitchell, 2006; Reich et al., 2008; Tiwana & McLean, 2005). This holds true for both physical and virtual teams (Cummings, Espinosa, & Pickering, 2009; Espinosa, Slaughter, Kraut, & Herbsleb, 2007). Successful expertise coordination leads to higher levels of productivity, satisfaction with the process, and satisfaction with the product (Andres & Zmud, 2002; Jiang et al., 2006). These results are achieved via better-quality teamwork (Hoegl & Gemuenden, 2001) and better resolution of goal conflicts (Sherif, Zmud, & Browne, 2006). Other studies have explored how shared knowledge and increased member capacity can enhance coordination (Dibbern, Winkler, & Heinzl, 2008; Espinosa et al., 2007).

In particular, expertise coordination is considered one key to ISD project success (Tiwana & McLean, 2005). Expertise coordination is the process of knowledge integration and the outcome of exchanging and combining knowledge through interactions among team members (Faraj & Sproull, 2000; Okhuysen, 2001; Reich & Benbasat, 1996). Although the term *knowledge integration* captures the transfer of knowledge between individuals, *expertise coordination* is the process of knowledge integration that facilitates a common understanding of project objectives and the approaches for reaching those objectives (Mitchell, 2006). It is the coordinated application of individually held expertise in the accomplishment of tasks at the project level. The integration of individuals' specialized expertise to perform ISD-related tasks is the essence of the team's capability (Grant, 1996).

The impact of coordination effectiveness and process implementation on a software development team's performance is so significant that it outweighs the impact of more traditional team factors (e.g., group resources, professional experience, and use of software development methods) (Faraj & Sproull, 2000). In the extant IS literature, however, findings regarding the relationship between coordination and ISD performance have been inconclusive.

According to variations of the theory of planned behavior, coordination behavior is dependent upon individuals' ability and willingness to coordinate. Past studies have adopted perceived behavioral control to indicate that individuals perform certain behaviors when they have a high level of ability to perform those behaviors (Ajzen, 2002). Studies of global software development teams and outsourcing have also indicated that members' abilities (such as client-specific knowledge, shared knowledge, or absorptive capacity) are critical for reducing coordination costs and improving coordination effectiveness (Dibbern et al., 2008; Espinosa et al., 2007). Expertise coordination ability refers to the extent to which team members are able to manage interdependencies. As an important antecedent to coordination behavior, the ability to coordinate expertise is the ability to manage information through the processes of exchange, integration, and utilization (Faraj & Sproull, 2000; Tiwana & McLean, 2005). Ability is a critical determinant of actual coordination behavior (Qu, Ji, & Nsakanda, 2012). For example, the effectiveness of knowledge integration is limited by the ability of two parties to exchange knowledge (Szulanski, 1996). In addition, expertise coordination willingness implies mutual consent among the team members to unify their efforts through the exchange and integration of knowledge and expertise (Gerwin, 2004). Team members' coordination willingness is one factor that determines their coordination behavior. In fact, a lack of willingness to exchange knowledge has been cited as one critical

factor that decreases the effectiveness of knowledge exchange (Hislop, 2003; Stenmark, 2000; Wiewiora, Murphy, Trigunarsyah, & Brown, 2014; Wiewiora, Trigunarsyah, Murphy, & Coffey, 2013).

Multiple dimensions of success must be considered in order to represent the perspectives of various stakeholders in the project (Schwalbe, 2009). In the team and group research streams, team performance and attitudinal perception have been used as two major measures of success (Cohen & Bailey, 1997). Team performance often indicates how efficiently and effectively the team arrives at the final outcome. A high-performance ISD team will meet or exceed the standard measures of scope, cost, and schedule for its project (Jiang, Motwani, & Margulis, 1997). System quality has also been used by many researchers to evaluate system success and project teamwork effectiveness (DeLone & McLean, 1992; Nidumolu, 1995, 1996a, 1996b; Pitt, Watson, & Kavan, 1995). Attitudinal perception has also been considered important to success (Mathieu, Maynard, Rapp, & Gilson, 2008; Muller & Turner, 2007). Team processes include interpersonal interaction and task performance, both of which result in personal perceptions of satisfaction. To summarize, measuring personal work satisfaction can capture individual attitudes, measuring software quality can capture the interests of users and the organization, and measuring project performance reflects the views of project management. Prior literature has considered these forms of success important in the evaluation of the development process (Aladwani, 2002; DeLone & McLean, 2003; Markus & Mao, 2004).

We conclude that coordination is an important factor influencing ISD project performance. However, the measures of coordination (i.e., coordination ability, coordination willingness, or coordination behavior) in previous research have varied. Thus, as we mentioned above, the inconclusive results regarding coordination and performance might be caused by inconsistent measures of coordination. Table 1 summarizes

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Quantitative Studies			
Author(s)	Hypotheses	Results	Concept/Construct
Hsu et al. (2012)	• Coordination (+) performance	• Supported	Behavior
Chen et al. (2009)	• Vertical coordination (+) project performance • Horizontal coordination (+) project performance • Horizontal coordination (+) closing perception gaps between users and IS developers	• Supported • Supported • Supported	Behavior
Cummings et al. (2009)	• Crossing spatial boundaries (+) coordination delay • Crossing temporal boundaries (+) coordination delay • Synchronous communication (–) coordination delay • Synchronous communication × cross spatial boundaries (–) coordination delay • Cross spatial boundaries × cross temporal boundaries × synchronous communication (–) coordination delay • Asynchronous communication (–) coordination delay • Cross spatial × cross temporal × asynchronous communication (+) coordination delay	• Supported • Supported • Supported • Not supported • Partially supported • Supported • Not supported	Behavior Coordination delay (Time lag in resolving issues, clarifying communication, and reworking tasks)
Kanawattanachai & Yoo (2007)	• Task–knowledge coordination (t1) (+) performance (t1) • Task–knowledge coordination (t1) (+) task–knowledge coordination (t2) • Task–knowledge coordination (t2) (+) performance (t2) • Task–knowledge coordination (t2) (+) task–knowledge coordination (t3) • Task–knowledge coordination (t3) (+) performance (t3)	• Not supported • Not supported • Supported • Not supported • Supported	Ability
Bharadwaj et al. (2007)	• Manufacturing–IS coordination (+) integrated IS capability • Manufacturing–marketing coordination (+) manufacturing performance • Manufacturing–supply chain coordination (+) manufacturing performance	• Supported • Supported • Supported	Behavior and ability
Jiang et al. (2006)	• Coordination (–) residual performance risk • Coordination (+) performance	• Supported • Supported	Behavior
Mitchell (2006)	• Internal knowledge integration (+) enterprise application integration (EAI) project completion	• Partially supported	Ability
Tiwana & McLean (2005)	• Expertise integration (+) creativity	• Supported	Behavior
Li et al. (2003)	• Horizontal coordination (+) IS–user climate • Horizontal coordination (+) IS–department satisfaction	• Supported • Not supported	Behavior
Andres & Zmud (2002)— <i>Experiment</i>	• Coordination strategies → project success (organic coordination > mechanistic coordination) • Interdependent × coordination strategies → project success • Goal conflict × coordination strategies → project success	• Supported • Partially supported • Not supported	Behavior (mechanistic and organic)

(continued)

Quantitative Studies			
Author(s)	Hypotheses	Results	Concept/Construct
Faraj & Sproull (2000)	- Expertise coordination process (+) team effectiveness - Expertise coordination process (+) team efficiency	- Partially supported - Partially supported	Willingness
Nidumolu (1996a)	Fit as mediation - Vertical coordination (+) process control - Vertical coordination (+) product flexibility - Horizontal coordination (+) process control - Horizontal coordination (+) product flexibility Risk-based model - Vertical coordination (–) software performance risks - Horizontal coordination (–) software performance risks - Horizontal coordination (+) product flexibility	- Not supported - Not supported - Not supported - Supported - Supported - Not supported - Not supported	Behavior
Nidumolu (1995)	Vertical coordination (–) project uncertainty Vertical coordination (–) residual performance risk Horizontal coordination (–) residual performance risk Horizontal coordination (–) project performance	- Supported - Supported - Not supported - Supported	Behavior
Qualitative Studies			
Author(s)	Relations Surfaced	Concept	
Sherif et al. (2006)	Coordination mechanisms (–) goal conflict - Monitoring - Communication - Rewarding	Ability	
Espinosa et al. (2007)	Shared knowledge (+) coordination Task awareness and presence awareness (+) coordination - Temporal coordination - Technical coordination - Process coordination	Ability	
Dibbern et al. (2008)	Vendor absorptive capacity (–) coordination costs Required client-specific knowledge (+) coordination costs Offshore-specific client–vendor distance (+) coordination costs	Behavior	

Table 1: Coordination studies in IS literature.

recent coordination studies in IS/IT that have not arrived at consistent results.

Research Model and Hypotheses

The purpose of this study is to examine the relationships among expertise coordination willingness, expertise coordination ability, and expertise coordination behavior, and their impacts on ISD project outcomes (project performance and system quality) and personal work satisfaction. Based upon the background discussion, we propose the research model shown in Figure 1.

When developing information systems, it is necessary to acquire and transfer

required knowledge among members of an ISD project team (Lee et al., 2015). Members who have both an awareness of and access to knowledge can avoid knowledge barriers that restrict knowledge exchange among members (Yli-Renko, Autio, & Sapienza, 2001). The effectiveness of coordination is limited when senders are unable to express knowledge and receivers are unable to absorb knowledge (Hinds & Pfeffer, 2003; Ko, Kirsch, & King, 2005). Beyond gaining access to knowledge, team members must also exchange and integrate knowledge during the ISD process. After becoming aware of who has critical knowledge, individuals exchange knowledge by relating,

encoding, communicating, decoding, and absorbing it. Therefore, in this study, *expertise coordination ability* is defined as the ability to process information through the knowledge exchange, integration, and combination. The level of knowledge exchange performance is reduced when the ability to coordinate is lacking (Faraj & Sproull, 2000; Qu et al., 2012). The effectiveness of knowledge exchange and integration is constrained by differences between the mental representations of the sender and of the receiver. Expertise cannot be exchanged among team members when the sender cannot express the information or the receiver cannot understand it, or when confusion occurs

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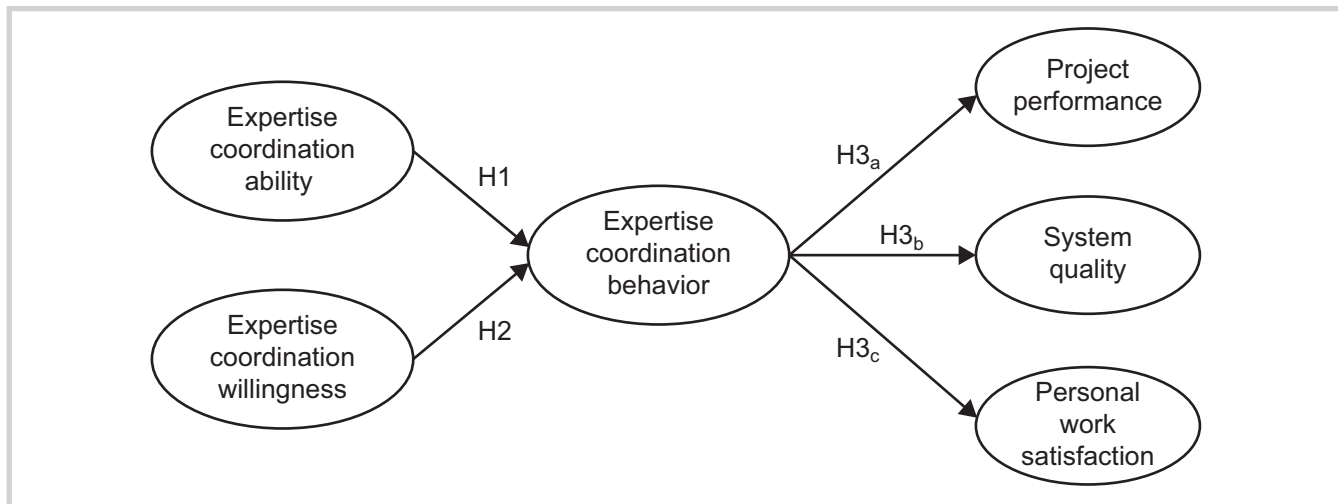


Figure 1: Proposed research model.

because there is no common interpretation (Szulanski, 1996).

Intention-behavior theories have also made similar assertions. According to the theory of planned behavior, behavior is determined by intention and ability (perceived controls) (Ajzen, 1991). Initially, perceived behavioral control referred to one's perceived resources and opportunities when conducting certain behavior. An individual's perception of or confidence in his or her ability has been referred to as *self-efficacy*, defined by Bandura (1986) as "people's judgments of their capabilities to organize and execute courses of action required to attain designated types of performance" (p. 391). Perception of one's ability is strongly correlated to both intention and behavior (Armitage & Conner, 2001). The chance of observing a certain behavior is reduced when the perception of the ability to perform that behavior is low because the success rate of performing the target action is low. Thus, we hypothesize the following:

H1: Expertise coordination ability is positively associated with expertise coordination behavior.

While cognitive limitations inhibit the ability to exchange knowledge, motivational limitations suppress the

willingness to integrate knowledge (Hinds & Pfeffer, 2003; Szulanski, 1996; Yli-Renko et al., 2001). The term *cognitive limitation* implies that experts may not estimate exactly what knowledge is needed by others who have less expertise. Thus, cognitive limitations reduce the experts' ability to deliver their knowledge, even if they are willing to exchange that knowledge. The term *motivational limitation* means that the motivating force reverses from encouragement to inhibition, under conditions such as competition. A system that induces people to compete for promotions, raises, or rewards no longer encourages cooperation; it effectively sets people against one another. Expertise is a competitive resource; thus, motivation limitations inhibit experts' willingness to exchange information and knowledge. *Willingness* refers to the extent to which team members intend to exchange knowledge with other team members, and to combine and utilize existing knowledge to solve problems. Previous research has indicated several reasons for low levels of willingness, such as fear of losing power, interpersonal relationships, lack of incentives, lack of confidence, the organizational climate, low reciprocity, conflict avoidance, uncertain rewards for sharing, and insufficient trust (Disterer, 2001; Ipe, 2003).

These inhibitors stop team members from coordinating expertise with other team members, reducing the likelihood that actual coordination behavior will be observed. A lack of willingness to exchange knowledge has been cited as one critical factor that erodes the effectiveness of knowledge exchange (Hislop, 2003; Stenmark, 2000; Wiewiora et al., 2013, 2014). In contrast, Wang and Noe (2010) reviewed 76 qualitative and quantitative studies published in different fields (e.g., management, organizational behavior, human resource development, applied psychology, and information systems), and the results consistently showed that knowledge sharing intention has a direct effect on knowledge sharing behavior. Ryan and O'Connor (2009) argued that knowledge sharing is a key process in ISD, especially with regard to expert/tacit knowledge. Expertise includes a significant amount of tacit knowledge (Stenmark, 2000), and previous studies have shown that willingness to coordinate knowledge has an important impact on the actual behavior exhibited (Inkpen & Tsang, 2005; Tiwana & McLean, 2002, 2005). Thus, we propose H2 as follows:

H2: Expertise coordination willingness is positively associated with expertise coordination behavior.

Organizational information processing theory provides a foundation for understanding the benefits of coordination. This theory suggests that uncertainty promotes information processing. Organizations can better cope with uncertainty by reducing the need for information processing and increasing their ability to process information (Hsu, Lin, Cheng, & Linden, 2012). In an ISD project, performance is eroded by uncertainty—the gap between the required information and the available information (Daft & Lengel, 1986). Sources of uncertainty include complex or non-routine tasks, unstable task environments, and interdependence (Tushman & Nadler, 1978). When tasks become more variable, unstable, and highly interdependent, information exchange is required to counter the resulting uncertainty in order to maintain performance (Galbraith, 1974). The natural complexity of an ISD project requires a team of two or more people who perform specific tasks or functions and who manage interdependencies to reach a valued goal (Crowston & Kammerer, 1998; Malone & Crowston, 1994). To obtain sufficient information and perform these tasks, individuals may need to exchange information with other team members within the project. Furthermore, interdependence reduces the degree to which activities can be preplanned and increases the need for members to adjust throughout the project (Van de Ven et al., 1976). Andres and Zmud (2002) adopted information processing theory and argued that coordination can be viewed as information exchange, which is essential to overcome uncertainty and interdependence, and is critical for better team outcomes.

Team performance is the level of efficiency and effectiveness with which the team carries out the tasks to achieve the final outcome (Cohen & Bailey, 1997). Team performance can be improved through effective expertise coordination (Faraj & Sproull, 2000; Kraut & Streeter, 1995). By sharing detailed information and specific design requirements, teams

can develop an IS more effectively and efficiently. In a project team, performance is not just a function of having the “right” expertise on the team. Rather, expertise must be coordinated among team members (Faraj & Sproull, 2000). With a high level of expertise coordination, team members are able to explore the causes of problems, understand and evaluate assumptions, and develop new solutions to improve current practices (Lubit, 2001). Coordinating the expertise possessed by individual team members can be viewed as a way to increase the project team’s information processing capability, which can help the team cope with the uncertainties caused by various conditions. Previous literature has identified a strong relationship between expertise integration and increased development efficiency (Tiwana & McLean, 2005; Tsai & Ghoshal, 1998; Yang, 2005). Therefore, we hypothesize as follows:

H3_a: Expertise coordination behavior is positively related to project performance.

System quality is commonly used to evaluate the product performance of a project team (Nidumolu, 1995; Wallace et al., 2004a). An ISD project requires all team members to contribute to the development and implementation process. Team members are especially required to integrate their knowledge and expertise to discover problems, diagnose the causes of those problems, and generate solutions—all of which require coordination among team members to achieve the desired output. Teams are more creative when diverse knowledge and viewpoints are integrated. Under such conditions, the team will have sufficient expertise to conduct experiments or attempt to improve the development process (Tiwana & McLean, 2005). A comprehensive view can be constructed by incorporating the diverse views of the team’s various members. Having a comprehensive understanding of potential problems is an important antecedent to improving system quality (Liang,

Jiang, Klein, & Liu, 2010). Therefore, supported by organizational information processing theory and the empirical studies above, we hypothesize as follows:

H3_b: Expertise coordination behavior is positively related to system quality.

The third coordinated teamwork outcome, attitudinal perception, refers to the extent to which members are personally satisfied with the results (Hackman & Morris, 1975). Team processes include interpersonal interaction and task performance. Both positive and negative consequences may emerge. Conflicts and disagreements may lead to negative affective consequences or loss of rewards. Administrative or explicit coordination increases team performance, and also increases members’ satisfaction regarding the development process and their personal sense of accomplishment or opportunities (Andrea, 2001; Faraj & Sproull, 2000; Kraut & Streeter, 1995). Team members can obtain other people’s expertise and learn from the integration process by observing how different expertise is gathered to generate solutions and solve problems, resulting in personal growth and the emergence of other opportunities. As a result, members tend to be more satisfied. Empirical studies have shown that high cooperation leads to member satisfaction (Hoegl & Gemuenden, 2001; Janz & Prasarnphanich, 2003). Thus, we expect the following:

H3_c: Expertise coordination behavior is positively related to personal work satisfaction.

Research Methods

Data Collection

The target respondents of this study are members of ISD project teams. Because coordination behavior is performed by two or more people, teams with more than five members were considered for further analysis. Teams were solicited via the authors’ personal social network

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in Taiwan. We first contacted a key individual on each team to introduce the purpose of this study and to obtain permission to access other team members.

Participating teams received a survey package that was hand-delivered to the team contact. The package was comprised of individual envelopes containing the survey instruments and instructions for each team member and a large collection envelope for the team contact. Team members were instructed to complete the survey and return it to the team contact in their sealed individual envelopes. Team member participation was voluntary. The contact person placed the sealed individual envelopes in the large envelope and returned it to the researchers. Team contacts who had not returned the survey after four weeks were contacted and reminded to complete the survey. A total of 140 teams from the sampling pool showed a willingness to participate in this study. Of the 110 returned survey packages, six were excluded because their valid respondents were less than 70% of the project team size. In sum, a total of 525 members from 104 teams were retained for analysis. Table 2 shows certain demographic aspects of the respondents, their organizations, and their projects.

Measures

All constructs were obtained from past research and were measured by multi-item scales. Because all respondents were located in Taiwan, the scales were translated into Chinese. The translation work was done by the researcher and validated by a PhD candidate who was not involved in this study and who was fluent in both English and Chinese. Finally, a Chinese-to-English back translation was performed to ensure that the meanings of the translated items were consistent with those of the originals. In addition, the Chinese version was validated by another researcher and two project managers with industry experience. Minor revisions were completed before the final

distribution. After this stage, to ensure content validity, the revised items were reviewed by researchers experienced in this field.

In this study, we assess project success by measuring project performance, system quality, and personal work satisfaction, because project success has been identified as having distinct forms (e.g., Shenhar, Dvir, Levy, & Maltz, 2001). *Project performance* refers to how efficiently and effectively a team can complete its required tasks. *Efficiency* means accomplishing the work within

a schedule and a budget, and *effectiveness* represents the quality of the work. Several studies have highlighted the differences between process efficiency and adhering to budget and schedule (e.g., Basten, Joosten, & Mellis, 2011; Gray, 2001), and researchers have also considered adhering to budget and schedule “process success” (Keil, Rai, & Liu, 2013). We adopted four items to measure project performance (Guinan, Coopridge, & Faraj, 1998; Jones & Harrison, 1996). *Personal work satisfaction* refers to team member perceptions of the teamwork

Variables	Categories	%	Variables	Categories	%
Gender	Male	67.2	Size of IS department	< = 10	22.5
	Female	32.2		11–52	35.1
	Missing	0.6		51–100	18.0
Job position	Dept. manager	5.1		101–500	18.9
	Project leader	12.0		> 500	2.7
	Programmer	48.2	Size of project team	Missing	2.8
	Systems analyst (SA)	15.7		< = 7	70.3
	Network analyst (NA)/Database analyst (DBA)	4.9		8–15	19.8
	Others	13.7		16–25	2.7
	Missing	0.4		> = 26	5.4
				Missing	1.8
Industry type	Manufacturer	52.9	Project duration (years)	< 1	64.9
	Finance	12.5		1–2	26.1
	Education	4.8		2–3	5.4
	Healthcare	11.5		3–5	0.9
	Others	18.3		> 6	1.8
				Missing	0.9
Project development method	Software development life cycle (SDLC)	22.1	Categories		Year
	Rapid application development (RAD)	5.8	Working experience	Work	5.47
	Prototyping	20.2		Software	4.76
	Agile	4.8		Current company	3.90
	Extreme programming (XP)	5.8	Total sample size: 104 teams; 525 people		
	Rational unified process (RUP)	1.9			
	Microsoft solutions framework (MSF)	1.0			
	Others	6.7			
	Missing	31.7			

Table 2: Demographic analysis.

processes. Three items adopted from Hoegl and Gemuenden (2001) were used to measure personal work satisfaction. *System quality* refers to the extent to which the system is stable and easy to use. Four items adopted from Wu and Wang (2006) were used to measure system quality. All constructs were measured with multiple items using five-point Likert scales.

Based on the discussion and arguments in previous sections of this article, *expertise coordination* has been defined as the knowledge integration process and the outcome of knowledge exchange and combination via interactions among team members (Faraj & Sproull, 2000; Okhuysen, 2001; Reich & Benbasat, 1996). However, there are no particular measurement items to measure expertise coordination behavior, willingness, and ability, as Mayer, Davis, and Schoorman (1995) suggested. For our purposes, *expertise coordination behavior* refers to the exchange, combination, and integration of individually held specialized expertise in the accomplishment of tasks at the project level. To assess expertise coordination behavior, we adapted items from Collins and Smith (2006) to measure members' behaviors regarding knowledge exchange and combination, and we adapted items from Tiwana and McLean (2005) to measure members' behaviors of expertise integration. *Expertise coordination willingness* refers to the extent to which project team members are *willing* to exchange, combine, and integrate their unique expertise or knowledge to solve a problem. To assess expertise coordination willingness, we adapted items from Collins and Smith (2006) to measure members' willingness to exchange and combine knowledge, and we adapted items from Tiwana and McLean (2005) to measure members' willingness to integrate expertise. *Expertise coordination ability* refers to the extent to which project team members are *able* to exchange, combine, and integrate knowledge or expertise to find solutions for the problems.

To assess expertise coordination ability, we adapted items from Collins and Smith (2006) to measure members' ability to exchange and combine knowledge exchange, and we adapted items from Tiwana and McLean (2005) to measure members' ability to integrate expertise.

To ensure content validity, the items were reviewed by researchers with ISD project management experience. Reliability and validity analysis also showed that all related values meet the statistical standard, indicating that the items for willingness and ability to coordinate and coordination behaviors can properly assess their corresponding constructs. These three constructs are also all valid team-level measurements, according to the definition provided by Klein and Kozlowski (2000). Expertise coordination behavior, expertise coordination willingness, and expertise coordination ability were measured with seven items using five-point Likert scales. The measurements items are shown in Table 3.

To control for exogenous effects in the research model, we included three control variables: project size, project duration, and project development method. First, past studies have suggested that project size affects the outcomes of software project development (e.g., Keil et al., 2013; Wallace, Keil, & Rai, 2004b). Second, project duration is expected to affect the process performance of a project (Keil et al., 2013). Lastly, because different project development methods may influence the extent to which team members work conjointly, we control for project development method. We included paths from the three control variables to the three project success variables (project performance, system quality, and personal work satisfaction).

Measure Validation

In this study, PLS-Graph Version 3.01 (Chin, 1994) was used to validate the measurement model and test the hypotheses via path analysis. Item reliability, convergent validity, and

discriminant validity were examined. Individual item reliability can be examined by observing the factor loading of each item. A high loading implies that the shared variance between the construct and its measurement is higher than the error variance (Hulland, 1999). Factor loadings higher than 0.7 can be viewed as having high reliability, and an item with a factor loading of less than 0.5 should be dropped. Loadings are shown in Table 3. Cross-factor loadings are shown in Appendix A.

Convergent validity should be tested when multiple indicators are used to measure one construct. It can be examined by item-total correlation (ITC), composite reliability, and average variance extracted by constructs (AVE) (Fornell & Larcker, 1981; Kerlinger, 1986). To obtain the required convergent validity, ITC should not be lower than 0.3, and composite reliability should be higher than 0.7. Moreover, an AVE of less than 0.5 means that the variance captured by the construct is less than the measurement error, and the validity of a single indicator and construct is questionable. ITC is shown in Table 3. Composite reliability and AVE values are shown in Table 4. All values meet the stated standards.

Discriminant validity focuses on the extent to which the measures of constructs are different from one another (Messick, 1980). To have the required validity, the correlation between pairs of constructs should be lower than 0.90, and the square root of AVE for each variable should be higher than the correlation coefficient to each remaining variable (Bagozzi, Yi, & Philips, 1991; Chin, 1988). Table 5 shows that these conditions were met and lists the descriptive values of the variables. Overall, the reliability and validity of the measures are ensured. The individual responses were aggregated to the team level for further analysis.

Because data were collected from individual members of each team before aggregating to the team level, there was a need to evaluate the agreement among

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Constructs	Items	Loadings	ITC	Source
Expertise coordination willingness	1. Willing to exchange and combine information, knowledge, and skills with others	0.72*	0.58	Collins & Smith (2006); Tiwana & McLean (2005)
	2. Willing to freely share hard-to-find knowledge or expertise with other members	0.75*	0.61	
	3. Willing to exchange and combine ideas with others to solve problems or create opportunities	0.81*	0.69	
	4. Willing to share my expertise with others to bring new projects of initiative to fruition	0.83*	0.72	
	5. Willing to span several areas of expertise to develop shared project concepts	0.74*	0.67	
	6. Willing to synthesize and integrate expertise with others at the project level	0.77*	0.71	
	7. Willing to blend new project-related knowledge with what they already know	0.71*	0.64	
Expertise coordination ability	1. Be able to exchange and combine information, knowledge, and skills with others	0.81*	0.73	Collins & Smith (2006); Tiwana & McLean (2005)
	2. Be able to freely share hard-to-find knowledge or expertise with other members	0.81*	0.73	
	3. Be able to exchange and combine ideas with others to solve problems or create opportunities	0.84*	0.77	
	4. Be able to share my expertise with others to bring new projects of initiative to fruition	0.83*	0.75	
	5. Be able to span several areas of expertise to develop shared project concepts	0.82*	0.76	
	6. Be able to synthesize and integrate expertise with others at the project level	0.82*	0.75	
	7. Be able to blend new project-related knowledge with what they already know	0.70*	0.62	
Expertise coordination behavior	1. Members of this team exchanged and combined information, knowledge, and skills with others	0.68*	0.57	Collins & Smith (2006); Tiwana & McLean (2005)
	2. Members of this team freely shared hard-to-find knowledge with other members	0.71*	0.59	
	3. Members of this team exchanged and combined ideas with others to solve problems or create opportunities	0.74*	0.62	
	4. Members of this team shared expertise with others to bring new projects of initiative to fruition	0.77*	0.66	
	5. Members of this team span several areas of expertise to develop shared project concepts	0.72*	0.60	
	6. Members of this team synthesized and integrated expertise with others at the project level	0.75*	0.63	
	7. Members of this team blended new project-related knowledge with what they already know	0.54*	0.54	
Project performance	1. Ability to meet project goals	0.81*	0.64	Guinan et al. (1998); Jones & Harrison (1996)
	2. Expected amount of work completed	0.86*	0.72	
	3. Adherence to schedule	0.84*	0.70	
	4. Adherence to budget	0.75*	0.59	
System quality	1. The developed system is easy to use	0.86*	0.72	Wu & Wang (2006)
	2. The developed system is user-friendly	0.83*	0.69	
	3. The developed system is stable	0.84*	0.71	
	4. Response time of the developed system is acceptable	0.81*	0.66	
Personal work satisfaction	1. I could draw a positive balance for myself overall	0.78*	0.41	Hoegl & Gemuenden (2001)
	2. I would like to do this type of collaborative work again	0.75*	0.37	
	3. Teamwork promotes me professionally	0.66*	0.32	

* Indicates significant at $p < 0.05$

Table 3: Factor loadings and item-total correlation.

Constructs	Composite Reliability (CR)	Average Variance Extracted (AVE)
Expertise coordination willingness	0.92	0.62
Expertise coordination ability	0.93	0.67
Expertise coordination behavior	0.88	0.52
Project performance	0.91	0.70
System quality	0.92	0.73
Personal work satisfaction	0.83	0.62

Table 4: Composite reliability and average variance extracted.

team members. We used the R_{wg} index to evaluate the appropriateness of aggregating data from the individual level to the team level. R_{wg} was proposed by James, Demaree, and Wolf (1984) to test for consensus or variation within a group unit. R_{wg} compares the variability of a given variable with the expected variance. R_{wg} ranges from 0 to 1. A high R_{wg} means there is high internal consistency for individuals in one group. Although values of 0.7 and higher are considered to represent a good level of consensus for a particular scale and team, there is no general set of guidelines regarding what constitutes a valid data set or how to handle individual cases, with the context and the number of cases meeting the criteria being important (Lance, Butts, & Michels, 2006). Less than 4% of the team scales fell below the threshold of 0.7. Most of these exceptions were in the individual satisfaction scale, and most were still above the other-source cutoff of 0.6. Overall, agreement within

the teams appears to be high, when set in the context of common practice (Lance et al., 2006). The R_{wg} values are shown in Appendix B.

Common method bias may inflate or deflate the strength of hypothesized relationships when both independent and dependent variables are simultaneously collected from the same respondents. Harman's single-factor (1976) analysis was conducted in order to ensure that common method bias was not an issue in our data. All 32 items listed in Table 3 were entered into the SPSS, and a factor analysis with varimax rotation was used to perform the test. The results indicated that exactly six factors were extracted, and the first factor explained only 31% of the variance; hence, common method bias is not a concern in this study.

Results

The results of hypothesis testing are presented in Figure 2. Expertise coordination ability was shown to have a

significant relationship with expertise coordination behavior ($\beta = 0.20$). Thus, H1 was supported. Additionally, expertise coordination willingness was shown to have a significant relationship with expertise coordination behavior ($\beta = 0.40$), supporting H2. The result is in alignment with past studies showing that the willingness to coordinate knowledge is more important than coordination capability in determining the extent of knowledge coordination (Roper & Crone, 2003). Expertise coordination behavior was shown to have a significant, positive relationship with project performance ($\beta = 0.46$), system quality ($\beta = 0.44$), and personal work satisfaction ($\beta = 0.65$). Project teams performing more expertise coordination behavior were more likely to report better project performance, system quality, and personal work satisfaction, thus supporting H3a, H3b, and H3c. The finding confirms previous research on coordination in software teams, which found that expertise coordination had a higher impact on team performance than traditional factors such as group resources and administrative coordination (Faraj & Sproull, 2000).

The variables in Figure 2 explain 26% of the variance in expertise coordination behavior with ability and willingness, 23% of the variance in project performance, 23% of the variance in system quality, and 41% of the variance in personal work satisfaction. Moreover, the control variables (project size,

Variables	Mean	Std. Dev.	M3	M4	Correlation Matrix					
					(1)	(2)	(3)	(4)	(5)	(6)
(1) Expertise coordination willingness	4.43	0.28	-0.43	-0.06	0.79					
(2) Expertise coordination ability	3.56	0.42	-0.16	-0.10	0.30	0.82				
(3) Expertise coordination behavior	3.67	0.30	-0.47	0.84	0.45	0.32	0.72			
(4) Project performance	4.04	0.39	-0.34	0.21	0.19	0.30	0.39	0.84		
(5) System quality	3.65	0.39	-0.17	0.18	0.10	0.25	0.42	0.58	0.85	
(6) Personal work satisfaction	4.00	0.35	-0.70	0.66	0.29	0.25	0.61	0.24	0.37	0.79

Note: Diagonal elements in the correlation of construct matrix are the square root of the average variance extracted.

Table 5: Descriptive statistics and latent variable correlations.

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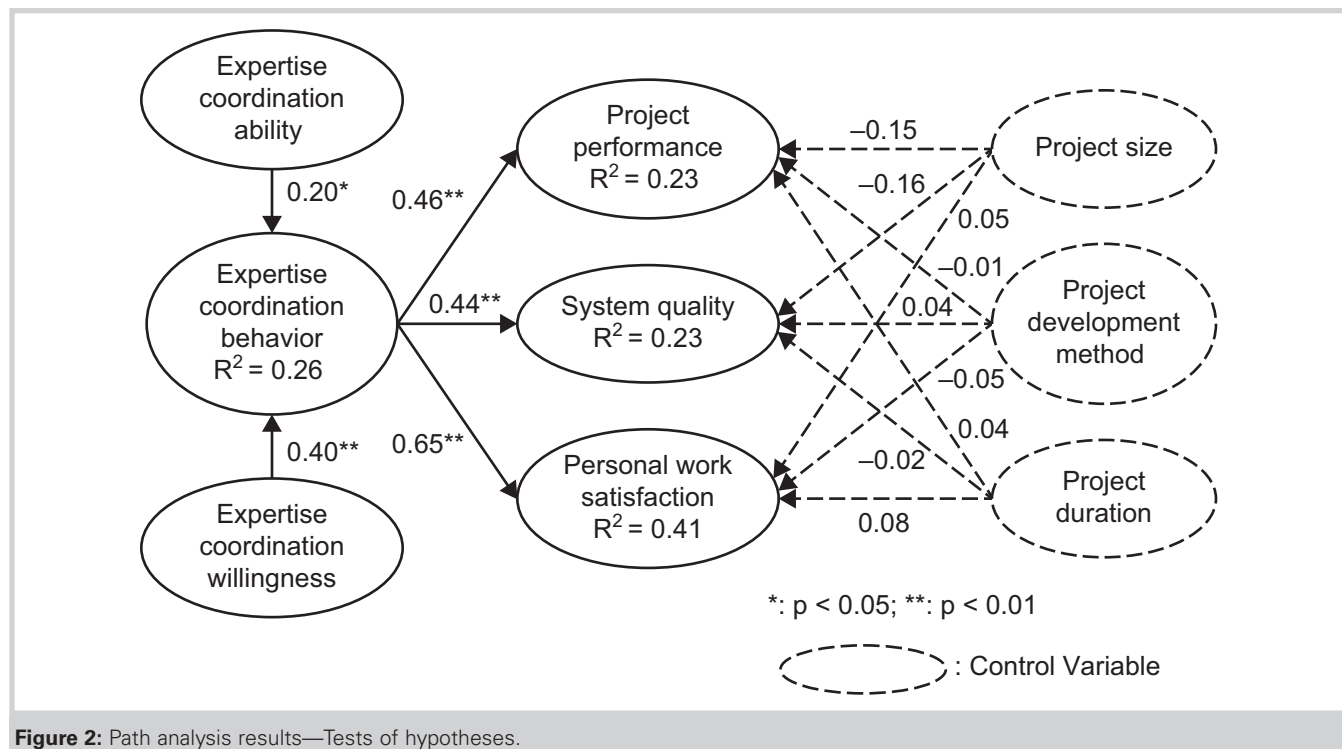


Figure 2: Path analysis results—Tests of hypotheses.

project duration, and project development method) have no significant effect on any of three dependent variables.

Because we suggested that behavior, willingness, and ability should be separated, and argued that coordination behavior transfers the impact of willingness and ability to project outcomes, the mediating effect of coordination behavior needed to be examined. Following Baron and Kenny's (1986) suggestion, we first performed separate tests on the direct effects of willingness and ability on project performance, system quality, and satisfaction, excluding expertise coordination behavior from the model. Although willingness had a direct impact on personal work satisfaction, ability had direct effects on project performance, system quality, and personal work satisfaction. The inconsistency of the direct effects of expertise coordination ability and expertise coordination willingness on project outcomes further highlighted the important role of expertise coordination behavior. We further examine the mediation role of

expertise coordination behavior, and the results are shown in Figure 3. Expertise coordination behavior was shown to have a significant relationship to three dependent variables. Only the path between ability and project performance remained significant. Furthermore, a Sobel test was conducted to evaluate the significance of the mediating effect. The results indicate that expertise coordination behavior fully mediates the relationship between willingness and three dependent variables (t -value = 2.78 for project performance, t -value = 3.08 for system quality, and t -value = 3.34 for satisfaction), and fully mediates the relationship between ability and both system quality and satisfaction (t -value = 2.05 for satisfaction, t -value = 1.99 for system quality, and t -value = 1.90 for project performance). Therefore, expertise coordination behavior fully mediates the relationship from willingness to all three dependent variables, as well as the relationship of ability to both system quality and personal work satisfaction. This finding is consistent with the theory

of planned behavior, which suggests that behavior is affected by both cognitive ability and willingness intention (Ajzen & Fishbein, 1980), and, as shown in our study, behavior then influences the project outcome.

Conclusion

An ISD project is a set of knowledge-intensive activities that require individuals with diversified backgrounds to contribute their unique knowledge to solve problems and carry out established procedures. Using a sampling of ISD teams, this study highlighted the importance of expertise coordination by showing its impact on multiple project outcomes. This study also found that coordination behavior is a function of both the willingness and the ability to coordinate, with behavior acting as a mediator in the relationships of willingness and ability to multiple outcomes.

Researchers may employ behavior in the determination of success when considering aspects of coordination.

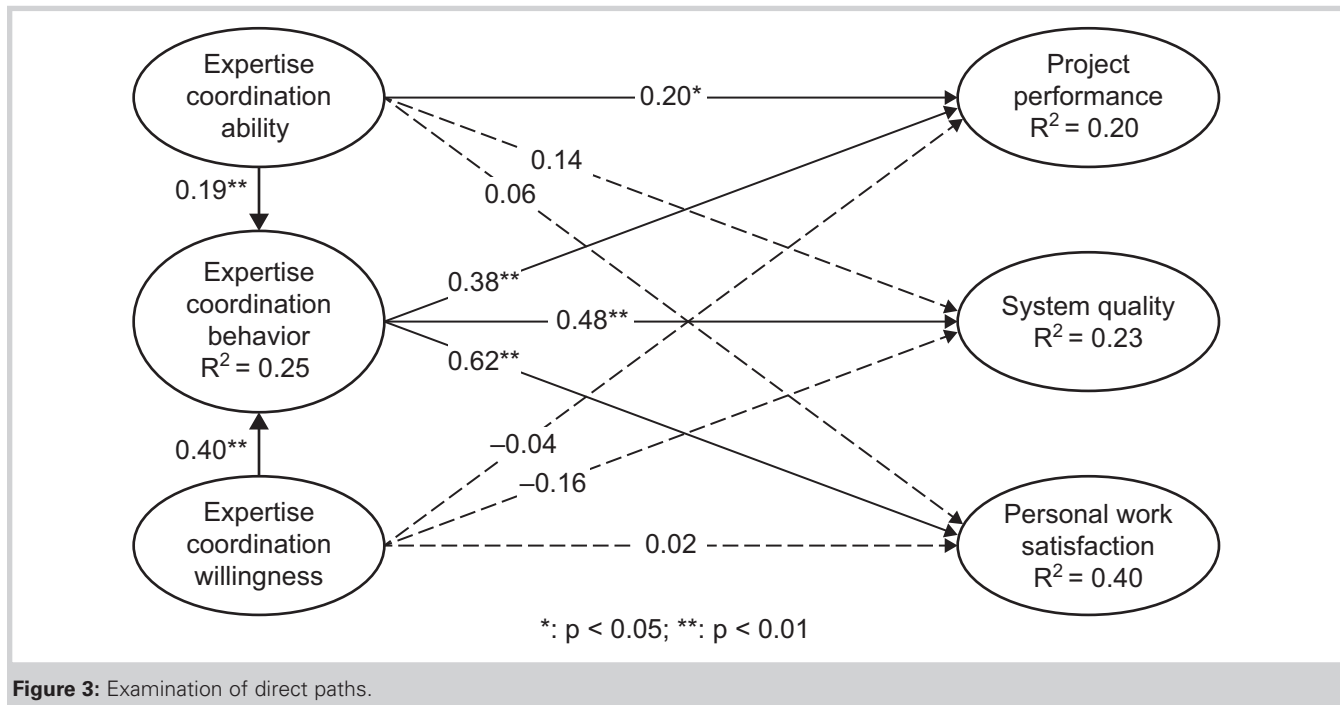


Figure 3: Examination of direct paths.

The structure employed in this study may serve to explain the inconsistent results of previous studies, which inconsistently represented the three components of coordination (ability, willingness, and behavior). From a more comprehensive view, coordination theories appear to hold in the IS context, revealing the potential for work on the appropriate components of coordination. Consequences flowing from behavior and management interventions (antecedents) should lead to willingness or ability.

The identification of different antecedents helps both researchers and practitioners determine what can be done to foster the willingness to coordinate and to enhance the ability to coordinate. Researchers may further clarify the picture by exploring potential antecedents for willingness and ability based on various organizational, behavioral, psychological, or small-group-based theories. For example, willingness to coordinate can be promoted by reducing interpersonal uncertainty, providing extrinsic motivations

to encourage appropriate actions, enhancing the intrinsic motivations of individual members, and exercising transformational leadership to improve team cohesiveness. Coordination ability can be enhanced through informative work communication, building standardized procedures for coordination, socialization, or providing adequate training to improve task skills and the ability to coordinate (Gerwin, 2004). Having a shared understanding of the tasks and knowing the location of expertise is also required for the efficient coordination of expertise (Mitchell, 2006; Nonaka, 1994).

Apart from contributing to the literature by disentangling the relation between willingness, ability, and behavior, this study has practical implications for practitioners. It suggests that although coordination behavior is important, coordination willingness is relatively more important than coordination ability. When coordination behaviors are low in teamwork, we recommend that project managers strategize to foster coordination

willingness. For example, project managers may plan team-building events to create a shared mental model. Team-building activities usually include giving and receiving mutual support, communication, and sharing. It can develop trust among team members and encourage collaboration, which can facilitate information processing within the team. As team members achieve higher levels of trust, the willingness to coordinate can be encouraged.

Moreover, coordination ability is another critical factor to enrich coordination behavior. Project managers not only can select team members with sufficient abilities when initializing projects but they also can provide some training courses to improve team members' skills and abilities. By improving both willingness and ability, members can coordinate their expertise and efforts to create project success.

This cross-sectional study is not without limitations. Coordination behavior may also impact the willingness and ability to coordinate in the future. For example, when members integrate their

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knowledge to solve problems successfully, members may generate more confidence and self-efficacy regarding conducting similar activities in the future. If conflicts or difficulties emerge during the coordination process, negative emotional consequences may be generated and may inhibit the willingness to coordinate in the future. Second, this study did not examine what could or should have been done to improve the willingness and ability to coordinate. Future research should explore potential antecedents, based on different theories. Third, we choose the cross-sectional survey as our research method because the purpose of a cross-sectional study is to find the prevalence of the result of interest (i.e., in this study, the relationships among expertise coordination willingness, abilities, behaviors, and project success) (Levin, 2006). Based on the results of our study, a future longitudinal study may further advance our understandings of the complicated expertise coordination willingness, abilities, and behaviors among team members during the ISD process. Finally, our data were taken from teams in Asia, limiting the generalizability of our findings.

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Dr. Jack Shih-Chieh Hsu is an Associate Professor at the Department of Information Management, National Sun Yat-sen University, Taiwan. He received his PhD in management information systems from the University of Central Florida, Orlando, Florida, USA. He has had articles published or accepted in *Information Systems Research*, *Decision Sciences*, *Information Systems Journal*, *Information & Management*, *Decision Support Systems*, and others. His research interests include various behavioral issues in electronic commerce, information security, and software project management areas. He can be contacted at jackshsu@mis.nsysu.edu.tw

Yu Wen Hung is a PhD candidate in Information Management at National Sun Yat-sen University, Taiwan. Her research focuses on behavioral issues in information systems development management and control and electronic commerce. Her research has been published in academic journals (e.g., *Information Systems Research*, *Information & Management*, *International Journal of Information Management*, *Project Management Journal*®) and in international conference proceedings (e.g., PACIS, ECIS, Pre-ICIS WISP). She can be contacted at d004020004@student.nsysu.edu.tw

Dr. Sheng-Pao Shih is an Associate Professor in the Department of Information Management at Tamkang University, Taiwan. He obtained his PhD in information management from National Central University, Taiwan and was a visiting scholar at the University of Central Florida. His work has been published in *Project Management Journal*®, *Information Systems Research*, *Information & Management*, *Computers in Human Behavior*, and others. His research interests include software project management, behavioral information security, and information technology personnel. He can be contacted at sbao@mail.tku.edu.tw

Dr. Hui-Mei Hsu is an Assistant Professor in the Business Management Department at National Kaohsiung Normal University, Taiwan. She received her PhD in management information systems from National Chung Cheng University, Taiwan. Her research interests include healthcare information systems, electronic commerce, healthcare management, and electronic medical records. Her published works have appeared in *Telemedicine and e-Health*, *Journal of Medical Systems*, and *Journal of Information Management*. She can be contacted at hmhsu@nknu.edu.tw

Expertise Coordination in Information Systems Development Projects

	Expertise Coordination Willingness	Expertise Coordination Ability	Expertise Coordination Behavior	Project Performance	System Quality	Personal Work Satisfaction
Will1	0.72	0.14	0.22	0.20	0.18	0.23
Will2	0.75	0.16	0.22	0.22	0.16	0.17
Will3	0.81	0.26	0.23	0.18	0.18	0.21
Will4	0.83	0.23	0.26	0.24	0.22	0.23
Will5	0.74	0.35	0.30	0.14	0.19	0.22
Will6	0.77	0.38	0.30	0.19	0.25	0.29
Will7	0.71	0.33	0.31	0.14	0.20	0.26
Ability1	0.24	0.81	0.11	0.21	0.22	0.32
Ability2	0.25	0.81	0.12	0.17	0.22	0.28
Ability3	0.31	0.84	0.14	0.19	0.21	0.27
Ability4	0.30	0.83	0.15	0.19	0.24	0.27
Ability5	0.28	0.82	0.22	0.17	0.22	0.29
Ability6	0.29	0.82	0.20	0.20	0.27	0.35
Ability7	0.29	0.70	0.23	0.19	0.24	0.29
Behavior1	0.28	0.20	0.68	0.29	0.35	0.35
Behavior2	0.21	0.13	0.71	0.23	0.30	0.32
Behavior3	0.20	0.20	0.74	0.37	0.33	0.36
Behavior4	0.23	0.14	0.77	0.37	0.38	0.40
Behavior5	0.21	0.09	0.72	0.34	0.30	0.30
Behavior6	0.28	0.11	0.75	0.38	0.38	0.40
Behavior7	0.14	0.15	0.54	0.21	0.29	0.36
PP1	0.20	0.19	0.36	0.81	0.42	0.36
PP2	0.20	0.20	0.40	0.86	0.43	0.41
PP4	0.19	0.18	0.38	0.84	0.49	0.47
PP5	0.21	0.17	0.33	0.75	0.46	0.34
SQ1	0.25	0.22	0.43	0.47	0.86	0.50
SQ2	0.23	0.16	0.41	0.43	0.83	0.49
SQ3	0.20	0.31	0.35	0.48	0.84	0.48
SQ4	0.18	0.25	0.34	0.45	0.81	0.45
Sat1	0.24	0.43	0.32	0.41	0.50	0.78
Sat2	0.27	0.18	0.41	0.42	0.47	0.75
Sat3	0.11	0.16	0.34	0.21	0.29	0.66

Appendix A: Cross-factor loadings.

#	will	ability	behav	pp	sq	sat
1	0.93	0.97	0.97	0.97	0.97	0.97
2	0.95	0.88	0.96	0.96	0.99	0.94
3	0.97	0.95	0.94	0.85	0.87	0.86
4	0.95	0.96	0.95	0.96	0.95	0.9
5	0.98	0.95	0.95	0.72	0.81	0.08
6	0.95	0.95	0.88	0.91	0.92	0.87
7	0.97	0.91	0.95	0.86	0.89	0.88
8	0.96	0.89	0.93	0.81	0.92	0.45
9	0.96	0.96	0.92	0.91	0.92	0.93
10	0.95	0.49	0.97	0.95	0.92	0.77
11	0.95	0.95	0.98	0.97	0.95	0.94
12	0.85	0.86	0.92	0.89	0.93	0.86
13	0.98	0.93	0.96	0.94	0.95	0.94
14	0.99	0.95	0.95	0.95	0.92	0.9
15	0.96	0.74	0.94	0.88	0.91	0.84
16	0.8	0.94	0.95	0.82	0.88	0.65
17	0.97	0.97	0.97	0.96	0.81	0.83
18	0.97	0.95	0.91	0.94	0.94	0.9
19	0.98	0.91	0.95	0.89	0.94	0.8
20	0.94	0.91	0.92	0.96	0.94	0.96
21	0.96	0.87	0.92	0.97	0.93	0.91
22	0.96	0.91	0.69	0.95	0.98	0.81
23	0.98	0.91	0.91	0.95	0.98	0.99
24	0.91	0.67	0.77	0.89	0.91	0.93
25	0.98	0.78	0.91	0.91	0.89	0.8
26	0.89	0.9	0.93	0.95	0.9	0.89
27	0.97	0.93	0.86	0.9	0.92	0.68
28	0.94	0.97	0.96	0.97	0.88	0.9
29	0.89	0.79	0.95	0.91	0.85	0.91
30	0.92	0.95	0.95	0.93	0.96	0.88
31	0.98	0.95	0.92	0.92	0.84	0.87
32	0.95	0.98	0.96	0.97	0.92	0.87
33	0.99	0.59	0.9	0.9	0.91	0.96
34	0.94	0.78	0.97	0.96	0.85	0.95
35	0.85	0.84	0.97	0.71	0.86	0.92
36	0.98	0.9	0.97	0.95	0.97	0.95
37	0.91	0.58	0.76	0.89	0.8	0.86
38	0.94	0.8	0.94	0.94	0.9	0.91
39	0.92	0.88	0.78	0.6	0.66	0.4
40	0.88	0.68	0.93	0.88	0.88	0.8
41	0.98	0.96	0.8	0.84	0.81	0.83

Appendix B: Inter-rater agreement (Rwg). *(continued)*

Expertise Coordination in Information Systems Development Projects

#	will	ability	behav	pp	sq	sat
42	0.96	0.92	0.98	0.93	0.85	0.77
43	0.93	0.86	0.99	0.95	0.97	0.83
44	0.84	0.89	0.88	0.72	0.79	0.78
45	0.99	0.87	0.91	0.89	0.85	0.77
46	0.99	0.98	0.94	0.91	0.87	0.95
47	0.99	0.71	0.98	0.94	0.92	0.92
48	0.98	0.93	0.92	0.82	0.96	0.81
49	0.97	0.88	0.78	0.56	0.82	0.61
50	0.98	0.9	0.91	0.96	0.97	0.94
51	0.98	0.87	0.96	0.96	0.94	0.98
52	0.92	0.92	0.94	0.92	0.95	0.92
53	0.96	0.95	0.98	0.95	0.83	0.81
54	0.97	0.6	0.97	0.93	0.96	0.8
55	0.96	0.95	0.95	0.9	0.97	0.84
56	0.97	0.93	0.96	0.89	0.85	0.92
57	0.93	0.96	0.95	0.95	0.97	0.94
58	0.91	0.87	0.8	0.63	0.82	0.45
59	0.96	0.82	0.91	0.85	0.96	0.87
60	0.96	0.83	0.93	0.91	0.79	0.88
61	0.82	0.89	0.9	0.89	0.84	0.9
62	0.96	0.94	0.86	0.92	0.94	0.67
63	0.99	0.96	0.95	0.91	0.89	0.9
64	0.98	0.92	0.96	0.95	0.89	0.94
65	0.97	0.98	0.95	0.96	0.93	0.97
66	0.94	0.94	0.97	0.88	0.91	0.72
67	0.98	0.9	0.96	1	0.97	0.95
68	0.99	0.92	0.93	0.91	0.95	0.67
69	0.92	0.94	0.97	0.89	0.96	0.89
70	0.97	0.93	0.96	0.98	0.94	0.92
71	0.97	0.94	0.87	0.79	0.94	0.81
72	0.98	0.96	0.93	0.71	0.96	0.95
73	0.96	0.94	0.95	0.91	0.81	0.91
74	0.98	0.94	0.93	0.94	0.88	0.77
75	0.96	0.96	0.99	0.92	0.92	0.97
76	0.98	0.92	0.93	0.95	0.86	0.95
77	0.99	0.88	0.94	0.9	0.95	0.94
78	0.96	0.95	0.95	0.98	0.97	0.91
79	0.97	0.85	0.9	0.94	0.91	0.94
80	0.9	0.94	0.94	0.87	0.92	0.62
81	0.99	0.96	0.55	0.65	0.81	0.87
82	0.98	0.92	0.96	0.89	0.95	0.95

Appendix B: Inter-rater agreement (Rwg). (continued)

#	will	ability	behav	pp	sq	sat
83	0.87	0.84	0.87	0.96	0.87	0.83
84	0.96	0.95	0.96	0.96	0.88	0.84
85	0.95	0.95	0.95	0.97	0.9	0.95
86	0.97	0.49	0.94	0.95	0.93	0.97
87	0.98	0.98	0.98	0.94	0.87	0.99
88	0.95	0.86	0.92	0.76	0.86	0.94
89	0.92	0.85	0.88	0.95	0.94	0.84
90	0.99	0.8	0.81	0.88	0.89	0.88
91	0.92	0.92	0.94	0.84	0.82	0.92
92	0.97	0.97	0.96	0.86	0.79	0.81
93	0.93	0.85	0.94	0.9	0.95	0.9
94	0.96	0.94	0.94	0.95	0.97	0.88
95	0.98	0.97	0.93	0.97	0.96	0.94
96	0.96	0.92	0.85	0.97	0.96	0.95
97	0.96	0.96	0.97	0.93	0.92	0.94
98	0.97	0.96	0.96	0.86	0.94	0.87
99	0.99	0.62	0.93	0.95	0.82	0.88
100	0.99	0.97	0.52	0.74	0.83	0.85
101	0.92	0.91	0.85	0.9	0.94	0.92
102	0.97	0.98	0.96	0.95	0.95	0.92
103	0.95	0.9	0.97	0.94	0.98	0.96
104	0.95	0.96	0.9	0.79	0.66	0.89

Note: will: Expertise coordination willingness; ability: Expertise coordination ability; behave: Expertise coordination behavior; pp: project performance; sq: system quality; sat: Personal work satisfaction

Appendix B: Inter-rater agreement (Rwg).